

## EXPERIMENT 24: PERFORM SHARPENING OF IMAGE USING HIGH-BOOST MASKS

### Aim

To sharpen an image using a High-Boost Mask in Python and OpenCV.

### Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image to grayscale.
4. Choose the high-boost factor A greater than 1.
5. Apply Gaussian blur to obtain the blurred image.
6. Compute the high-boost image using  $g(x,y) = A f(x,y) - \tilde{f}(x,y)$ .
7. Clip the pixel values to the range 0 to 255.
8. Display and save the sharpened image.

### High-Boost Mask

```
0  -1  0
-1  A+4  -1
0  -1  0
```

For A = 2:

```
0  -1  0
-1  6  -1
0  -1  0
```

### Python Code

```
import cv2
import numpy as np

# Read the input image
img = cv2.imread("input.jpg")

if img is None:
```

```
print("Image not found")
exit()

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# High-boost factor
A = 2.0

# Blur the image
blurred = cv2.GaussianBlur(gray, (5, 5), 0)

# Apply High-Boost Masking
high_boost = cv2.addWeighted(gray, A, blurred, -1, 0)

# Keep pixel values within valid range
high_boost = np.clip(high_boost, 0, 255).astype(np.uint8)

# Save the output image
cv2.imwrite("high_boost_output.jpg", high_boost)

# Display images
cv2.imshow("Original Image", gray)
cv2.imshow("High-Boost Sharpened Image", high_boost)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

## **Input**



## Output



## Result

Thus, the image was successfully sharpened using the High-Boost Mask technique with a high-boost factor  $A = 2$ .